

Pratham Khanna

Contact No: (+91) 89249 22903
Email ID: prathamkhanna.er2@gmail.com

GitHub Link: [GitHub](#)
LinkedIn: [LinkedIn](#)
Portfolio Link: [Portfolio](#)

ACADEMIC DETAILS

B.Tech. (Computer Science and Engineering)	2022-2026	MMMUT, Gorakhpur	8.32/10.0
Class XII (C.B.S.E.)	2022	H.A.L. Vidyalaya, Kanpur	96.2% (School Topper)
Class X (I.C.S.E.)	2020	Methodist High School, Kanpur	95.0%

INTERNSHIP/EXPERIENCE

Summer Intern ([View Certificate](#)) **Hindustan Aeronautics Limited (H.A.L)** *June, 2025 – July, 2025*

- RogiTrita.AI (Automated Medical Diagnostics and Prescription System)**
 - Project Link:** ([Repository Link](#)) ([Deployed Link](#)) ([Live Demo Link](#))
 - An **end-to-end, Agentic-AI** powered medical web app enabling users to submit symptoms and receive **AI-assisted diagnostic reports**.
 - Implemented a **multi-agent system** using **CrewAI** and **DeepSeek LLM** to process symptoms, medications, and lab data **asynchronously**.
 - Developed an **interactive** doctor dashboard enabling **100% review control**, and feedback integration before prescription generation.
 - Automated PDF report creation and integrated **secure cloud storage** with Cloudinary for easy access and **1-click** download.
 - Technologies Used:** **Python, Streamlit, MongoDB Atlas, Cloudinary, CrewAI, DeepSeek LLM, PDF generation (FPDF), Tavily.**

AI and Analytics Intern **Beautiful Planet.AI** *January, 2025 – June, 2025*

- Lawyer Assistance Workflow (Multi AI agent team + Retrieval Augmented Generation)**
 - Developed an **Agentic-AI** pipeline using **LangChain, CrewAI**, and **RAG** to automate **end-to-end** legal case preparation, from document input to draft generation.
 - Extracted structured text from scanned PDFs using **OCR**, enabling **digitization of legacy law documents**, across varied scan quality.
 - Integrated **vector database (ChromaDB)** for efficient **semantic search** and **retrieval** of legal codes, case references and precedent judgements.
 - Integrated **DeepSeek R1** model via **OpenAI-compatible, LiteLLM** interface to enable cost-efficient, high-performance inference.
 - Incorporated CrewAI's **role** and **goal-based agent architecture** for **100% modular** and **robust prompting**.
 - Technologies Used:** **Python, LangChain, CrewAI, DeepSeek LLM, ChromaDB, RAG, LiteLLM, LMStudio.**
- Biomass Estimation System (Computer Vision + Object Detection)**
 - Fine-tuned a **YOLOv5** model for tree stem detection using **10k+ augmented images**, enabling **accurate biomass estimation** with robust performance across diverse forest conditions through advanced **preprocessing** and **hyperparameter tuning**.
- Disease Classification System (Image Classification + Model Deployment)**
 - Fine-tuned a **ResNet-9** model for **36-class** leaf disease detection with **97%+ accuracy** and converted it to **ONNX** for fast, cross-platform deployment in **agricultural monitoring system**.

PROJECT(s)

Comp-Intell (Agentic AI Competitor Intelligence Team)

- Project Link:** ([Repository Link](#)) ([Deployed Link](#))
- Built an **AI-powered** system to help businesses analyze up to **5 competitors** per input by **scraping web content** and **processing internal company files** using LLMs and RAG.
- Integrated **DeepSeek LLM** to generate **market strategy** and **SWOT** analysis, improving output relevance by **30%** when internal documents were included.
- Used **Firecrawl** to extract structured data from competitor websites and **Exa AI** to discover relevant competitor URLs based on company URL or description, reducing manual research time by **>80%**.
- Implemented **Retrieval-Augmented Generation (RAG)** using **LangChain, FAISS**, and **HuggingFace embeddings** to deeply analyze company's internal files (PDF, DOCX, TXT).
- Tested on **10+ real-world** companies across domains (eco-lighting, SaaS), consistently delivering actionable intelligence with **>90%** user-rated usefulness.
- Technologies Used:** **Python, Streamlit, Firecrawl API, Exa API, LangChain, FAISS, DEEPSEEK LLM, HuggingFace Embeddings.**

ACHIEVEMENTS

- NPTel **Introduction to Machine Learning** course (2025) top **1% topper** among 13644 candidates. ([View Credential](#))
- Smart India Hackathon 2023. ([View Credential](#))
- Solved 150+ D.S.A. problems on Geeks for Geeks. ([View Profile](#))
- 5-star (Java) coding badge at Hacker Rank. ([View Profile](#))

SKILLS

Technical: Python, Java, n8n, LangChain, LangFlow, LangGraph, LlamaIndex, LangSmith, CrewAI, SciKit-Learn, TensorFlow, PyTorch, Keras, MySQL, MongoDB, Git, GitHub.

Curriculum: Data Structures and Algorithms, Machine Learning, Object Oriented Programming, Software Engineering.

Certificates: [Machine Learning Specialization](#) | [Generative AI with Large Language Models](#) | [LangChain for LLM Application Development](#) | [LangChain Chat with Your Data](#) | [Multi AI Agent Systems with CrewAI](#) | [Practical Multi AI Agents and Advanced Use Cases with CrewAI](#) | [I.I.T. Madras B.S. degree Foundational Level](#).

Interests: ML, NLP, Agentic AI, Image Processing, Speech Recognition, AI Automation, Retrieval Augmented Generation, LLMs, MCP servers.

Soft Skills: Problem solving approach, Hardcore Learner, avoid repeating mistakes, Analytical Thinking, Result Oriented, Flexibility, Trustworthiness, Honesty, Integrity, Ability to work in team environment.

POSITION(s) OF RESPONSIBILITY

- AI/ML Head** at Computer Engineering Society, MMMUT *May 2025 - ongoing*
- Executive Board Member** at Computer Engineering Society, MMMUT *June 2023 – May 2025*